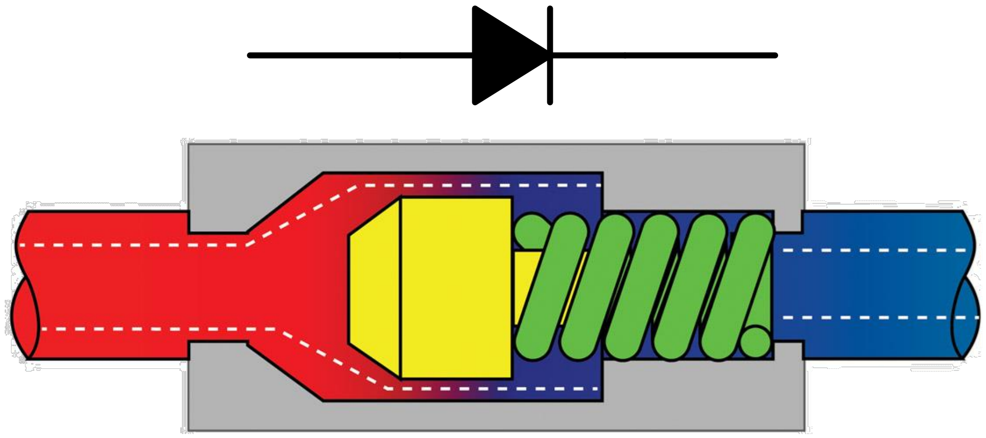
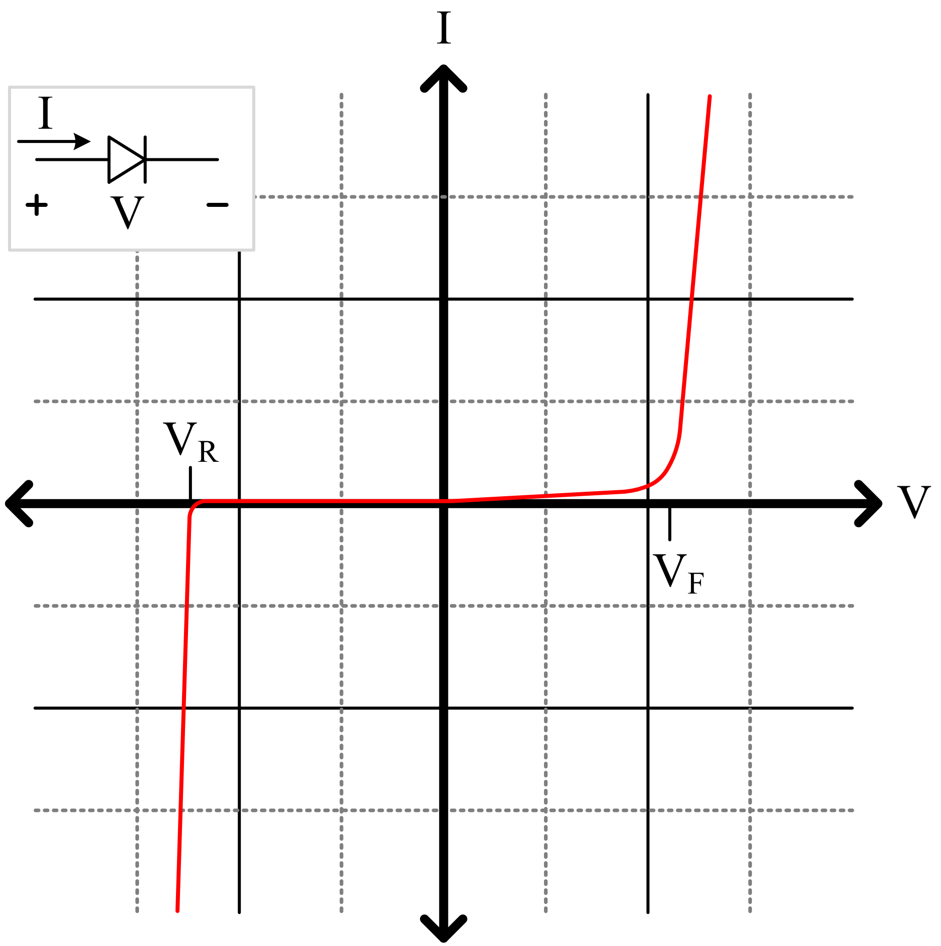
2.3 – Diodes

2.3.1 – Diode Operation and Circuit Model

Diodes are Like One-way Valves

- No flow until you push hard enough
- Blocks reverse flow… until it breaks!
- Like a check valve



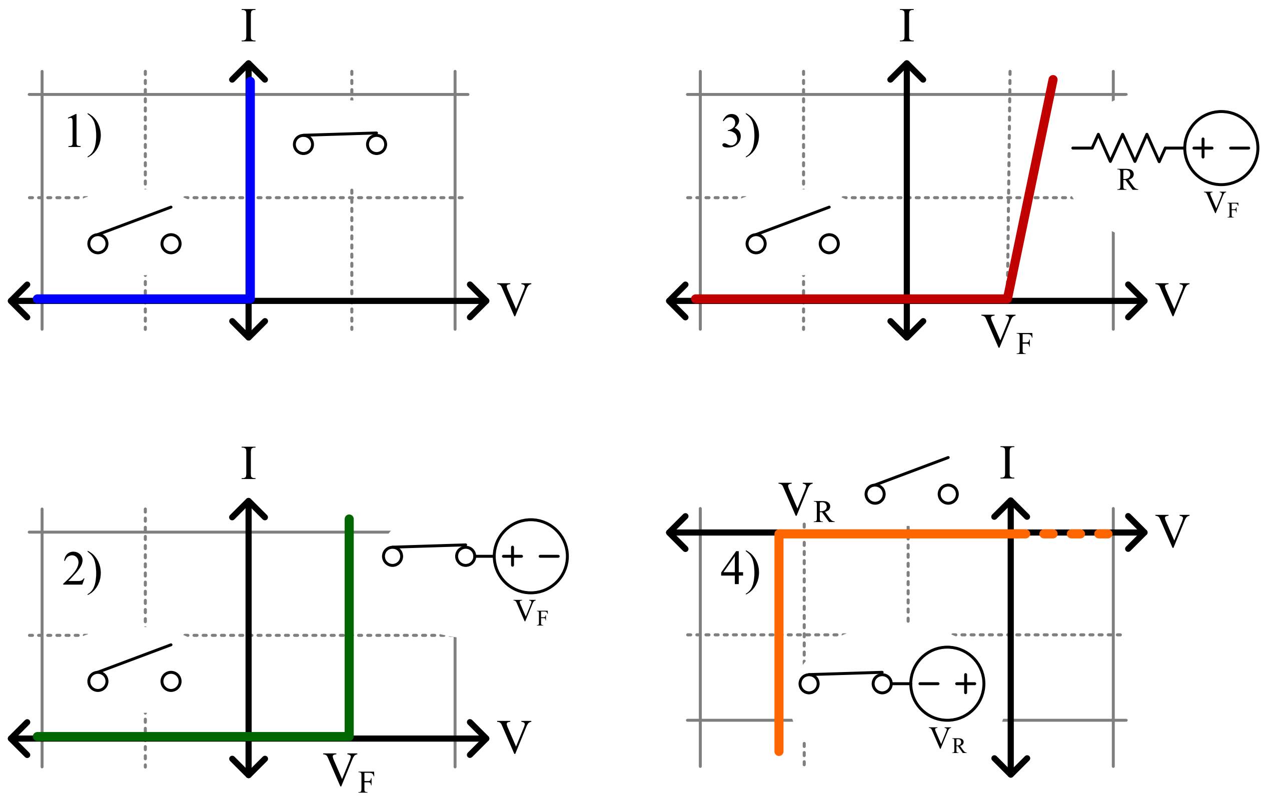


https://youtu.be/7ukDKVHnac4?si=s24e4jpI8bClEqjW&t=45

Diode Models

Rather than full exponential model, use ON/OFF model:

1) Simple switch
2) Switch with an offset voltage
3) Switch with offset voltage and resistance
4) Include reverse breakdown

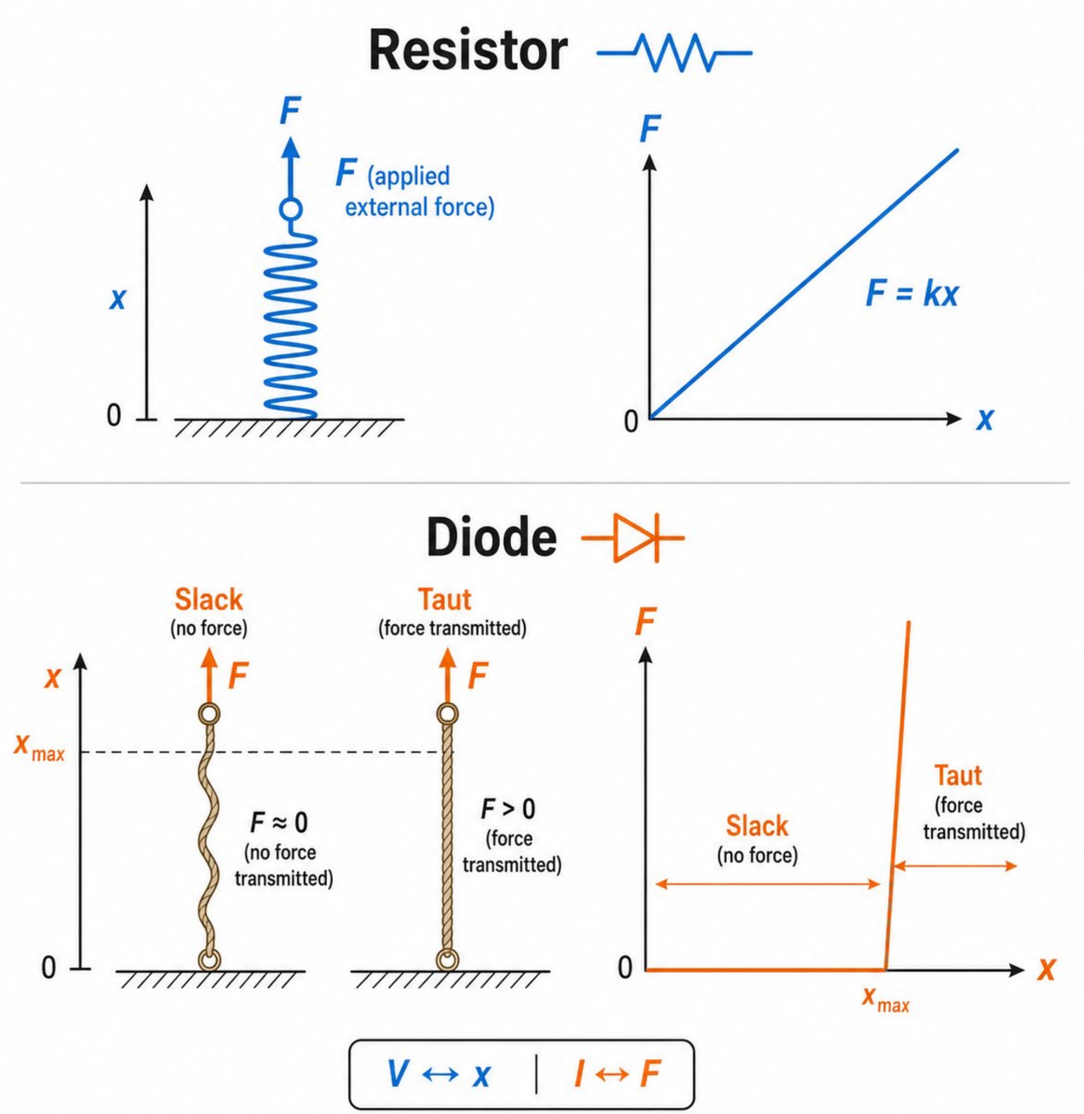


Diode Analogy

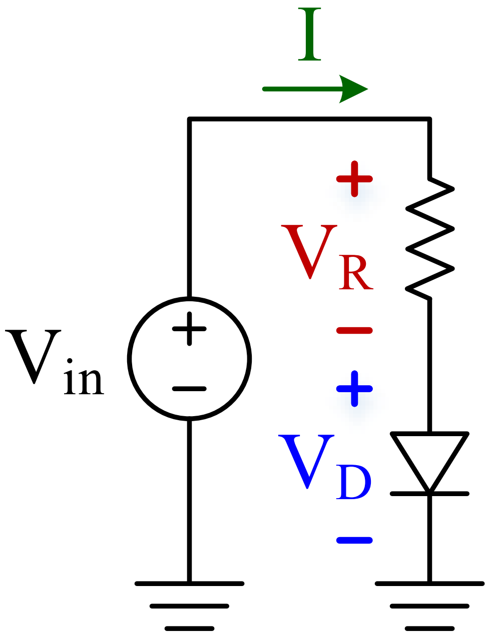
- Think about voltage V as a vertical position x
- Think about current I as an applied force F

- A **resistor** is like a **spring**:
  - Position and force are proportional

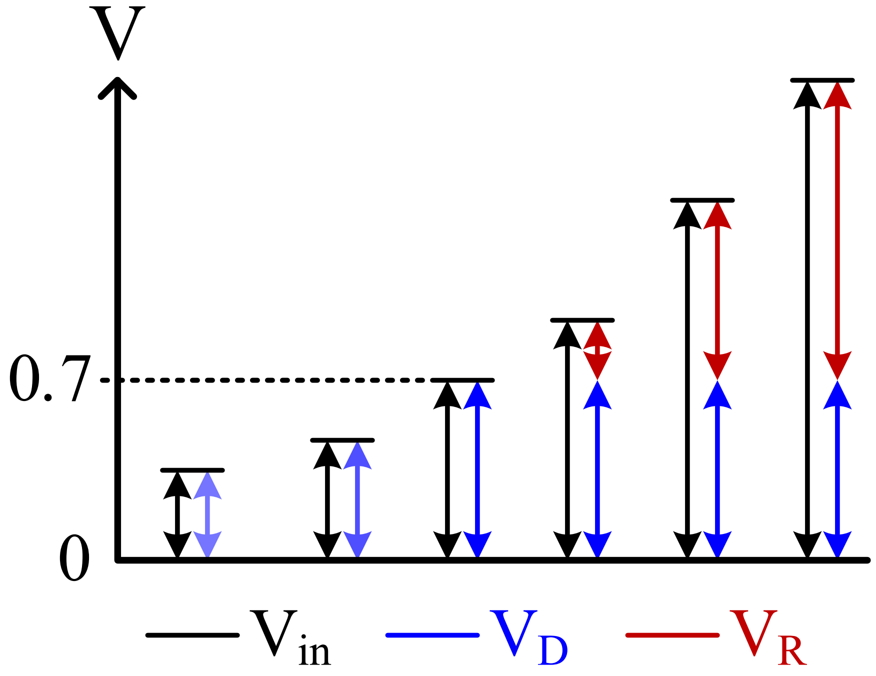
- A **diode** is like a **rope**:
  - No force until max position
  - Then, position mostly fixed for any force



Diode Circuit Behaviour – Example 1

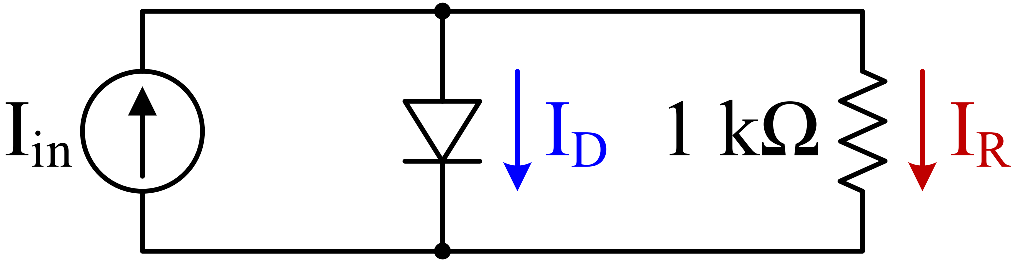


This diode turns ON at 0.7 V
Called V_F: Forward Voltage

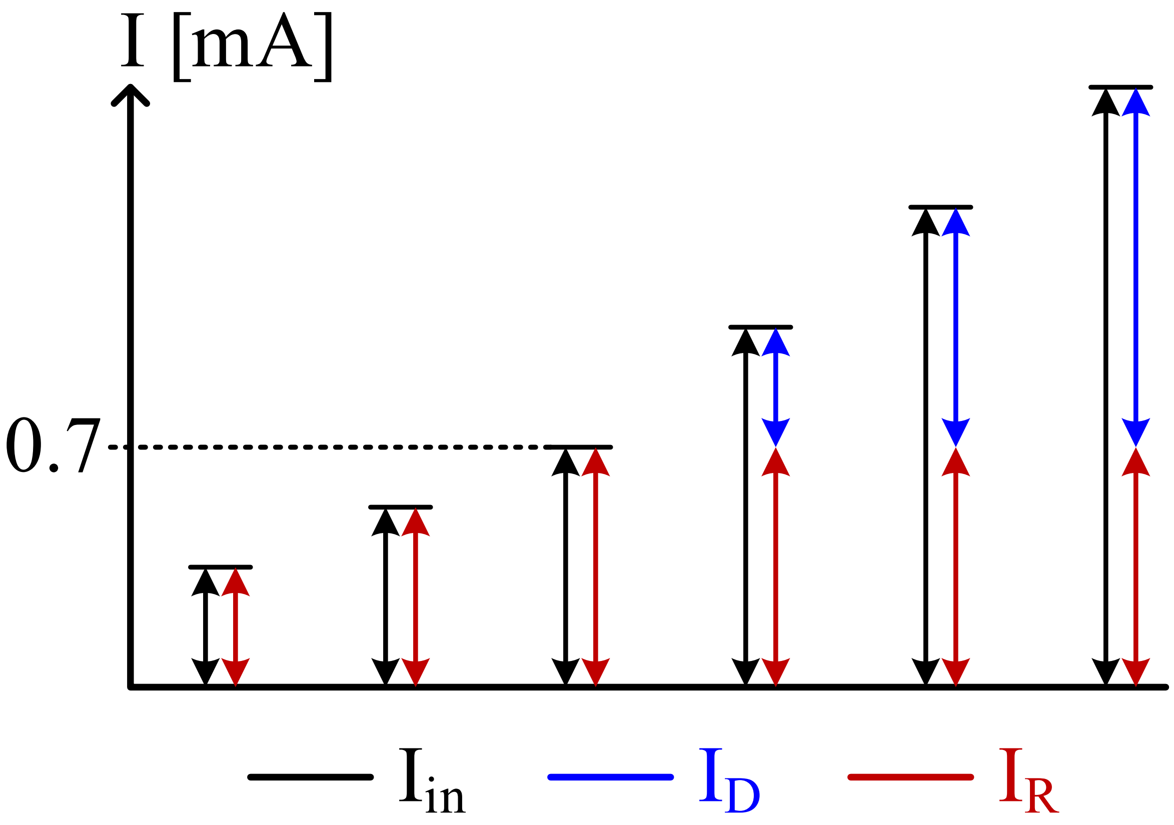


https://everycircuit.com/circuit/5715452701179904

Diode Circuit Behaviour – Example 2



This diode turns ON at 0.7 V
(when $I_{in} \times R = 0.7$ V)



https://everycircuit.com/circuit/5219441801560064

2.3.2 – Diode Types

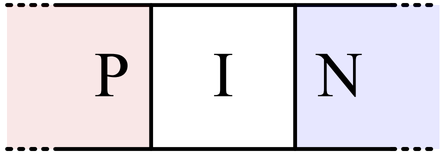
Diode Types: PN Junction

- “Normal” diode. E.g., pervasive 1N4148 diode:

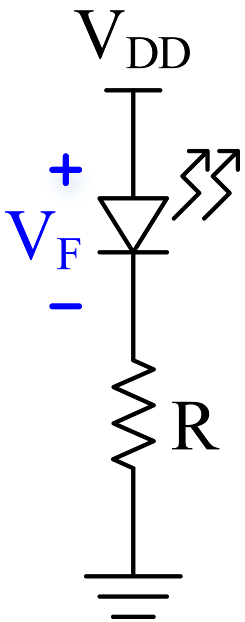
ELECTRICAL CHARACTERISTICS (Values are at T_A = 25°C unless otherwise noted) (Note 2)

Symbol	Parameter		Conditions	Min	Max	Unit
V_R	Breakdown Voltage		I_R = 100 μA	100		V
			I_R = 5.0 μA	75		V
V_F	Forward Voltage	914B / 4448	I_F = 5.0 mA	0.62	0.72	V
		916B	I_F = 5.0 mA	0.63	0.73	V
		914 / 916 / 4148	I_F = 10 mA		1.0	V
		914A / 916A	I_F = 20 mA		1.0	V
		916B	I_F = 20 mA		1.0	V
		914B / 4448	I_F = 100 mA		1.0	V
I_R	Reverse Leakage		V_R = 20 V		0.025	μA
			V_R = 20 V, T_A = 150°C		50	μA
			V_R = 75 V		5.0	μA
C_T	Total Capacitance	916/916A/916B/4448	V_R = 0, f = 1.0 MHz		2.0	pF
		914/914A/914B/4148	V_R = 0, f = 1.0 MHz		4.0	pF
t_{rr}	Reverse Recovery Time		I_F = 10 mA, V_R = 6.0 V (600 mA) I_{rr} = 1.0 mA, R_L = 100 Ω		4.0	ns

Diode Types: Light Emitting Diodes



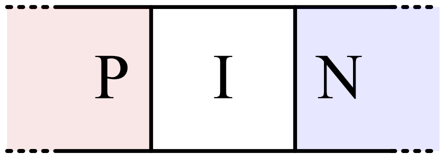
- LED brightness proportional to forward current, I_F
  - 20 mA max typical setpoint
- V_F dependent on LED colour and is poorly controlled
  - Sometimes ±0.5 V
- R sets current by dropping remaining voltage
  - $V_R = V_{DD} - V_F$, and $I_F = V_R/R$
  - Add enough voltage overhead to deal with V_F uncertainty
- Closed-loop feedback can provide a precise current independent of V_F (Transconductance Amplifier – OTA)

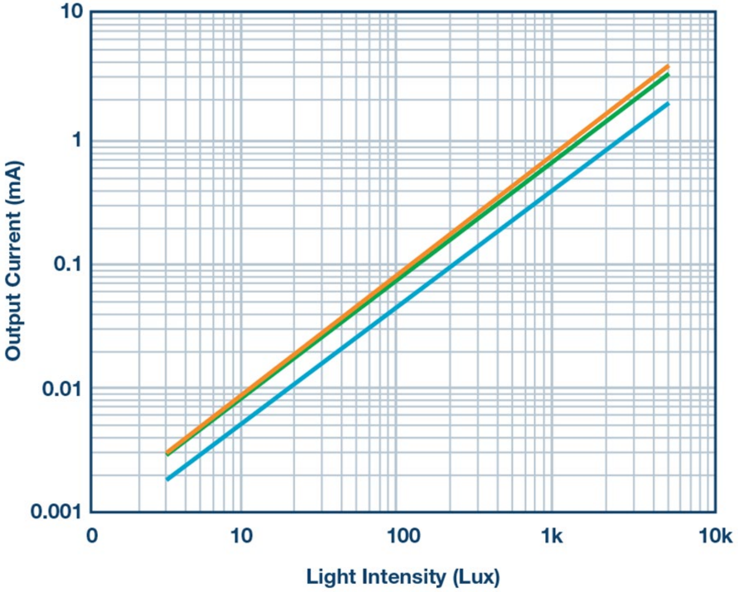


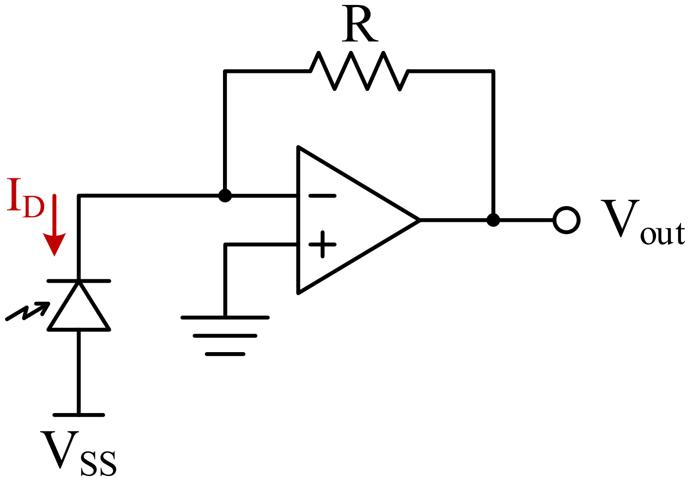
https://everycircuit.com/circuit/6713432258379776

Diode Types: Photodiode

- "Opposite" of an LED
- Allows reverse current flow when photons hit the depletion region
- Reverse current (I_D) proportional to illuminance:





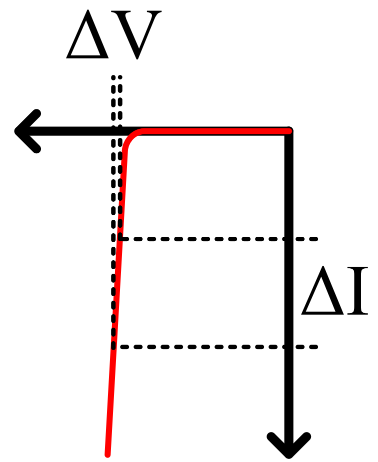


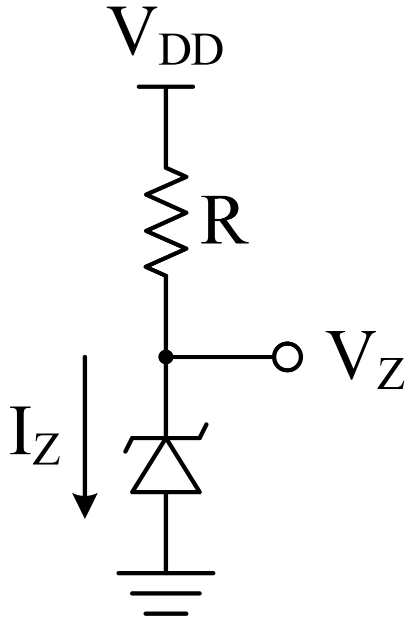
https://www.analog.com/en/resources/analog-dialogue/raqs/raq-issue-166.html

 https://everycircuit.com/circuit/5525277169680384

Diode Types: Zener Diodes

- V_F is poorly controlled
- V_R can be well-controlled → Use as a voltage reference
- Zener diodes operate on the edge of reverse breakdown to provide a reasonably precise voltage reference
  - Given a specified I_Z, a Zener diode will drop V_Z across it
  - Choose a precision resistor, R, to drop the rest of the voltage
- Looks “backwards” – it is because we want to use V_R, not V_F
- Analyze just like a regular diode, using V_Z instead of V_F



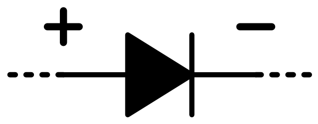


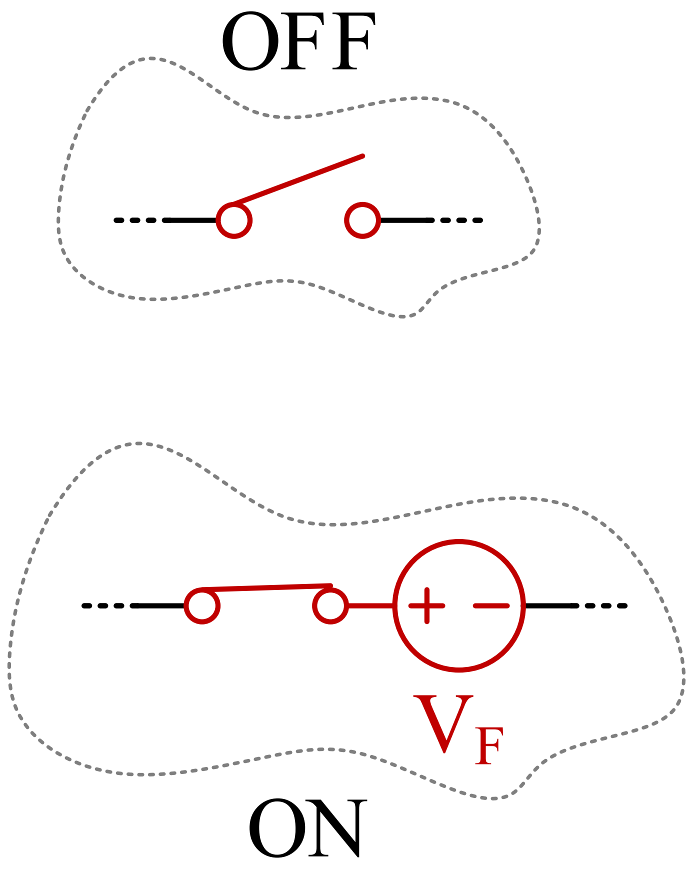
https://everycircuit.com/circuit/4975930875445248

2.3.3 – Diode Analysis Techniques

Diode Circuit Analysis – ON/OFF Model

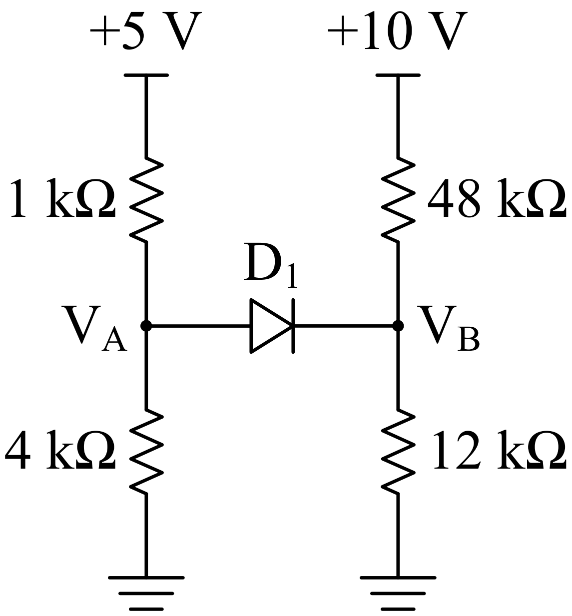
- Use two models: OFF & ON
- Draw circuit with diode OFF
  - Solve for V_D and any other unknowns
- Draw circuit with diode ON
  - Solve for V_D and any other unknowns
- Use OFF model when $V_D < V_F$ and ON model when $V_D \geq V_F$

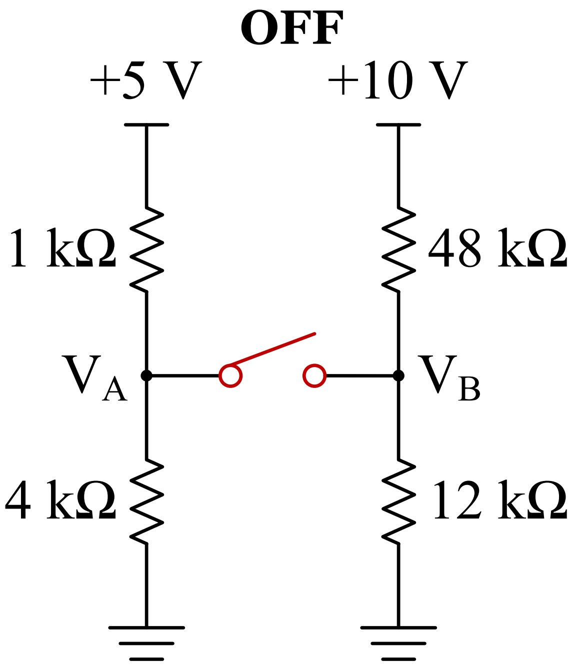




Diode Circuit Analysis Example

- Determine V_A and V_B. Use a basic diode model that does not conduct any current until 0.7 V is applied, at which point it conducts any amount of current.

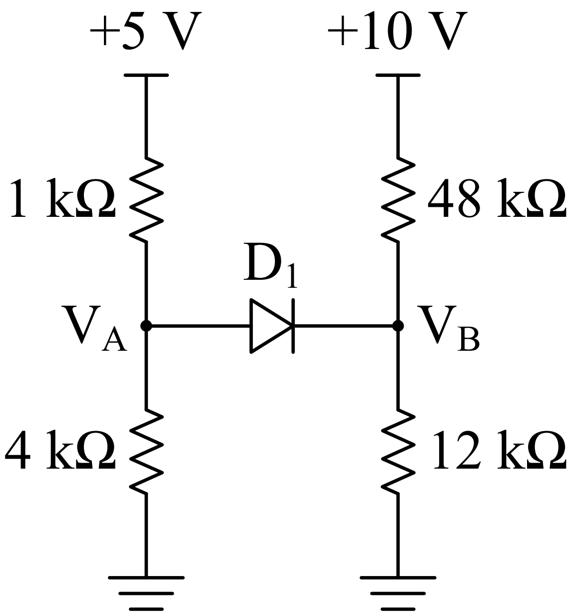


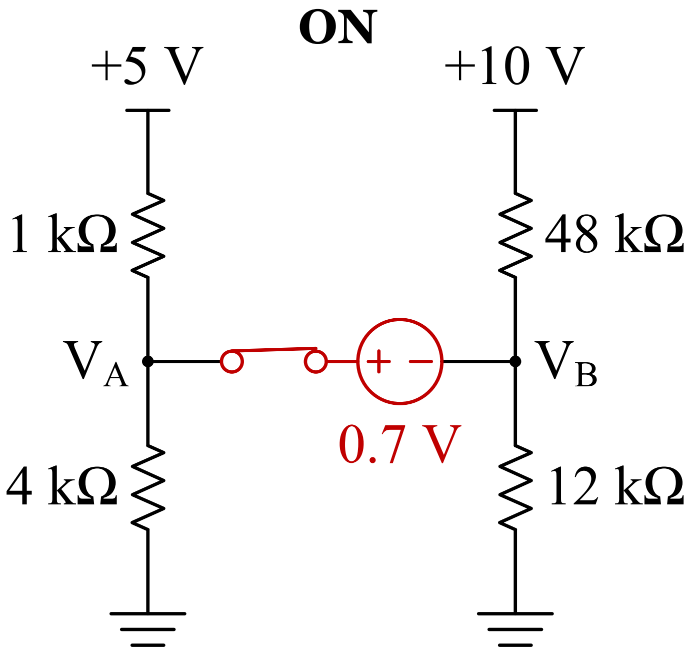


 https://everycircuit.com/circuit/5815933628841984

Diode Circuit Analysis Example

- Determine V_A and V_B. Use a basic diode model that does not conduct any current until 0.7 V is applied, at which point it conducts any amount of current.

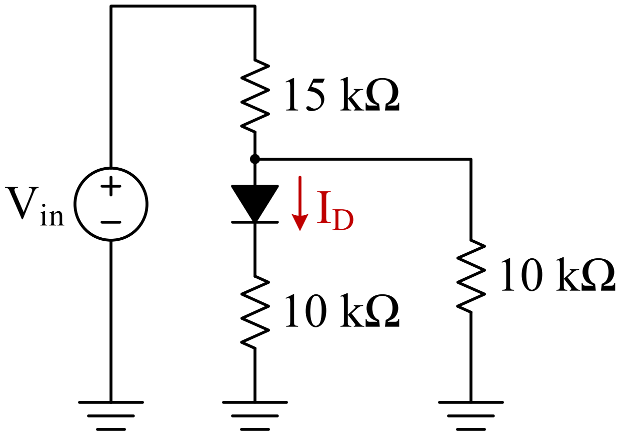




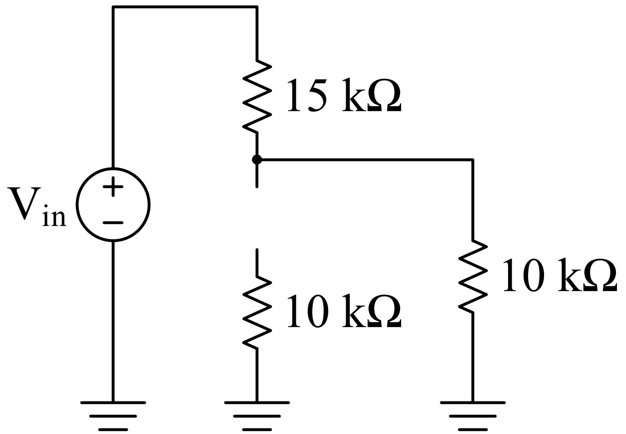
 https://everycircuit.com/circuit/5815933628841984

Diode Circuit Analysis Example: DC Sweep

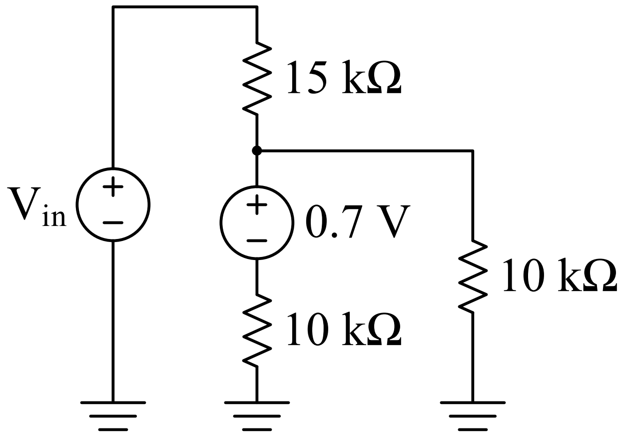
- Plot V_{D+}, V_{D-}, V_D, and I_D for V_{in} from 0 to 5 V. Use the same basic diode model.



OFF



ON



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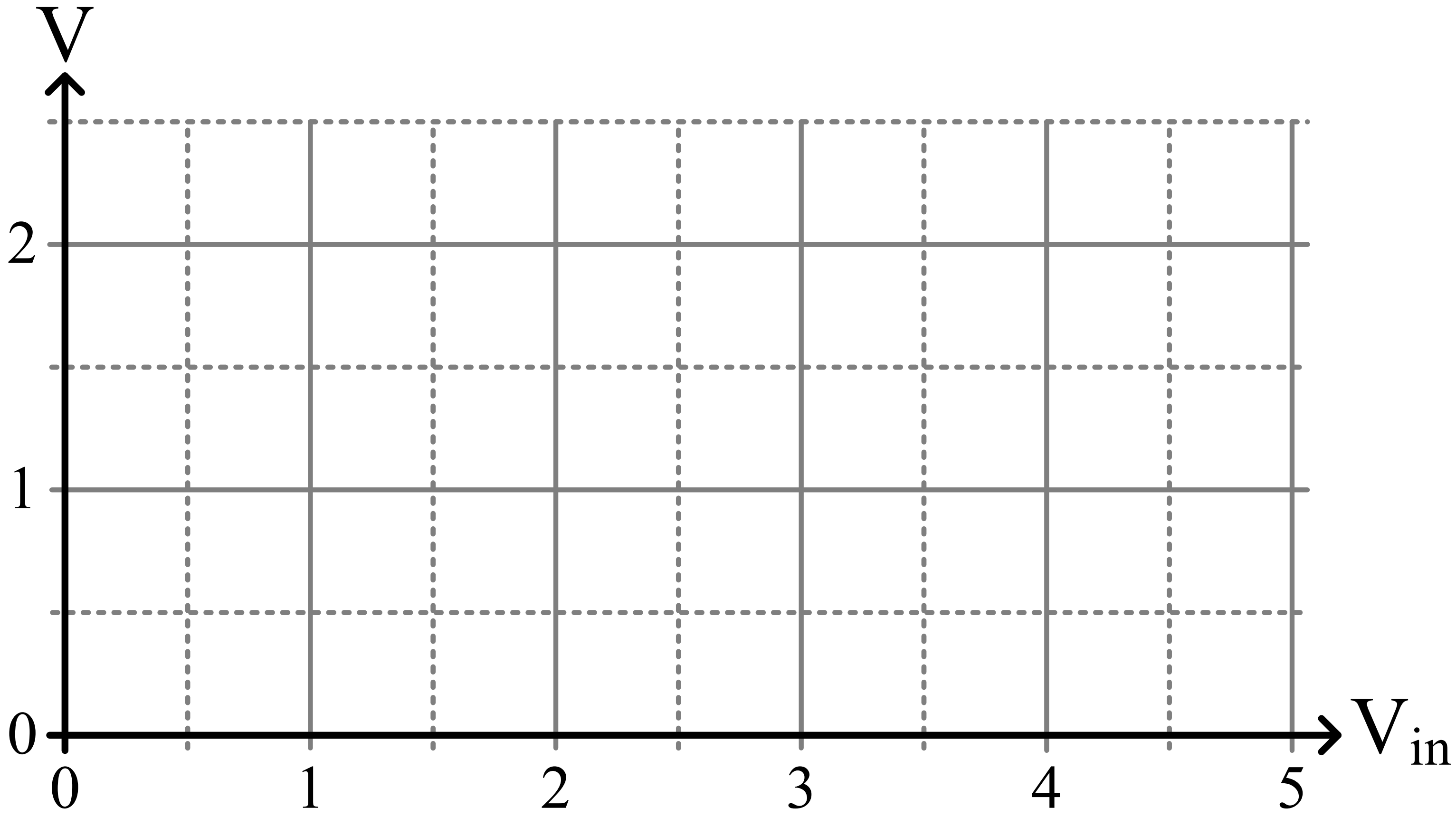
Diode Circuit Analysis Example: DC Sweep



https://everycircuit.com/circuit/4505789096132608

2.3.4 – Diode Circuits

Rectifiers

1. One that rectifies: a rectifier of many wrongs.

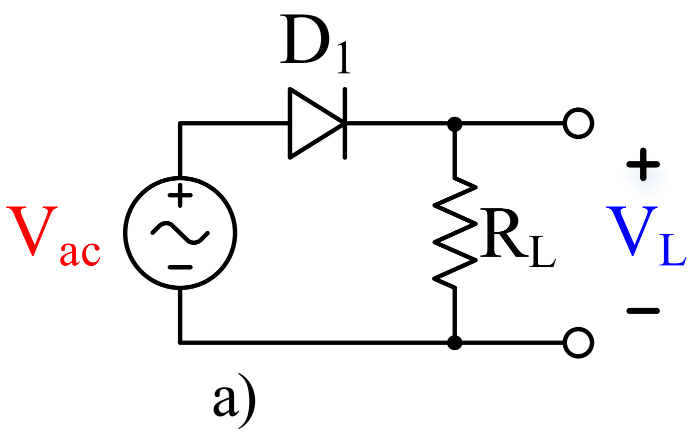
2. Electronics: A device, such as a diode, that converts alternating current to direct current.

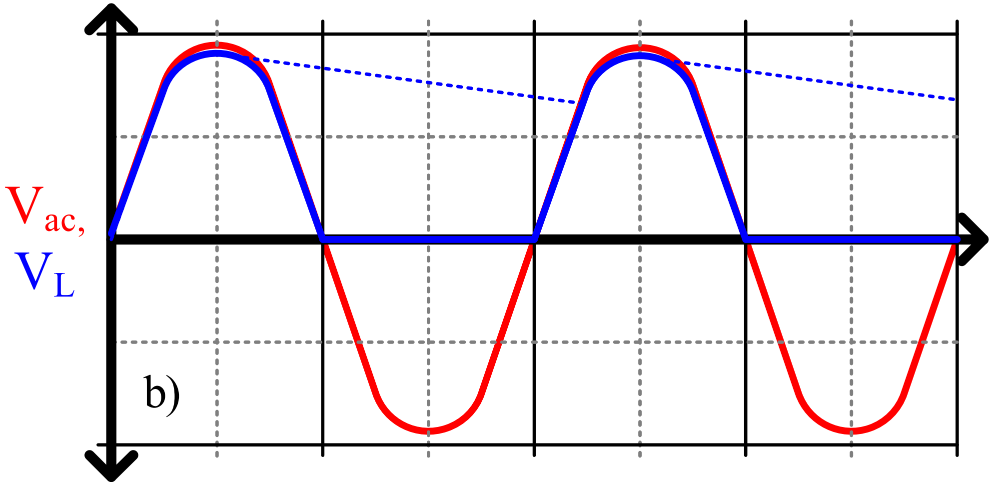
3. A worker who blends or dilutes whiskey or other alcoholic beverages.

ChatGPT's best efforts in 2024

Diode Circuits: Half-Wave Rectifier

- You can turn AC into DC with diodes
- Half-wave rectifier only lets the positive part of the AC cycle through
- Can add a large capacitance to smooth it out (Bulk Decoupling Capacitor)
    - Reduces the ripple

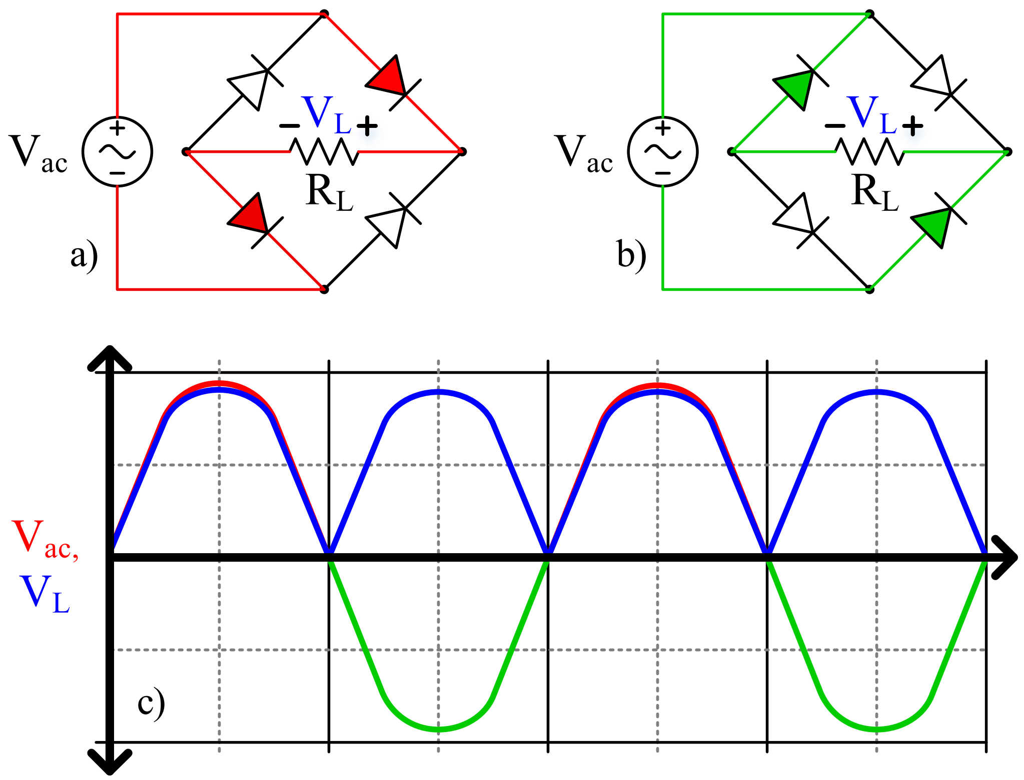




 https://everycircuit.com/circuit/6280785438703616

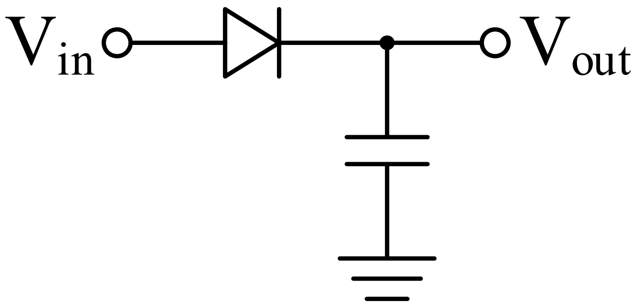
Diode Circuits: Full-Wave Rectifier

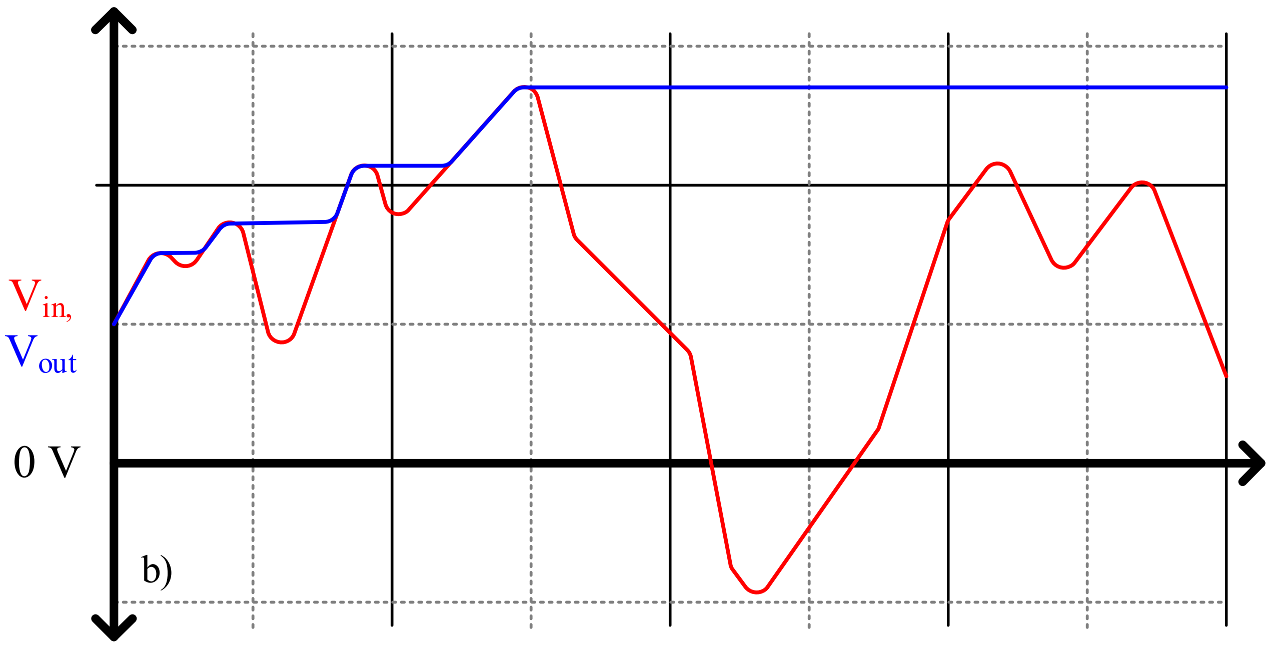
- A full-wave rectifier does a better job
- Also called **bridge rectifier**
- Two diodes ON for positive cycle
  Two diodes ON for negative cycle
- Routes current through the load in the same direction both times
- Again, can add a big capacitor to reduce the ripple



https://everycircuit.com/circuit/6031759485566976

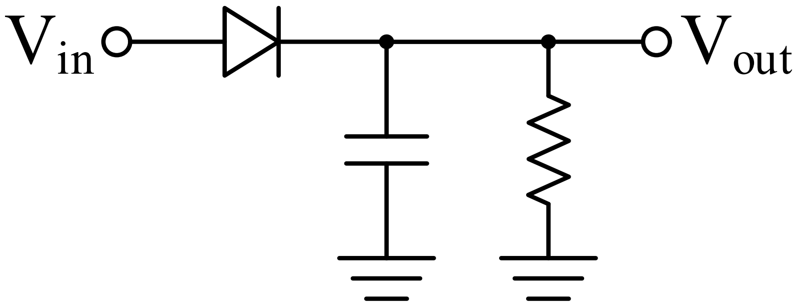
Peak Detector

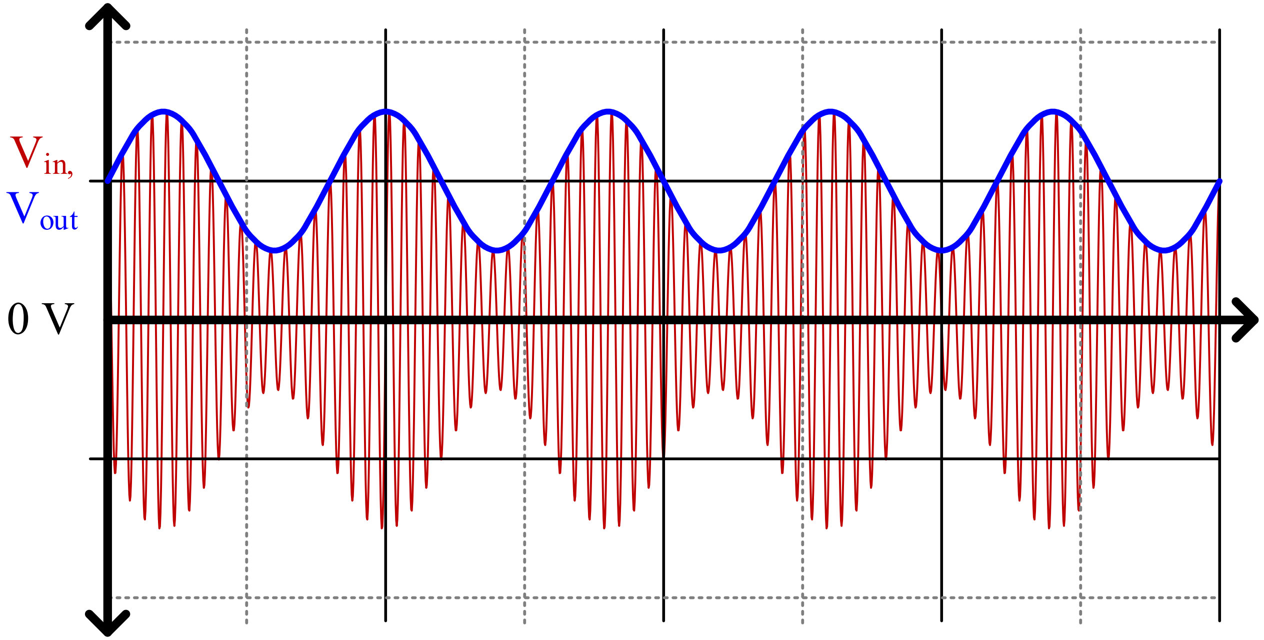




 https://everycircuit.com/circuit/6706811834728448

Envelope Detector

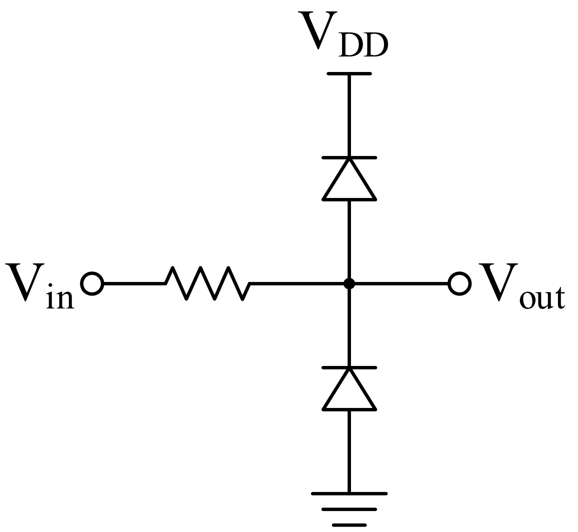


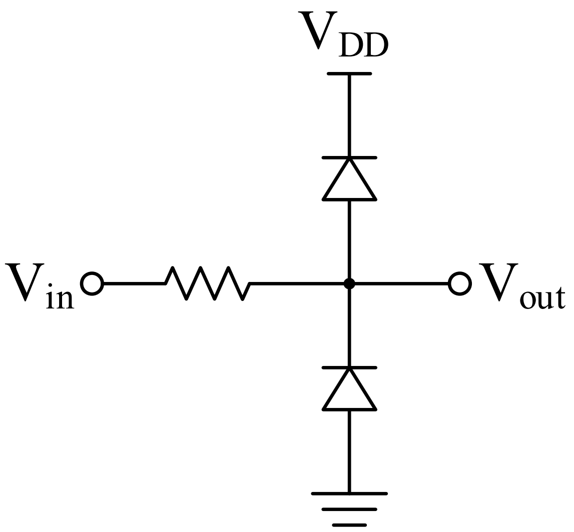


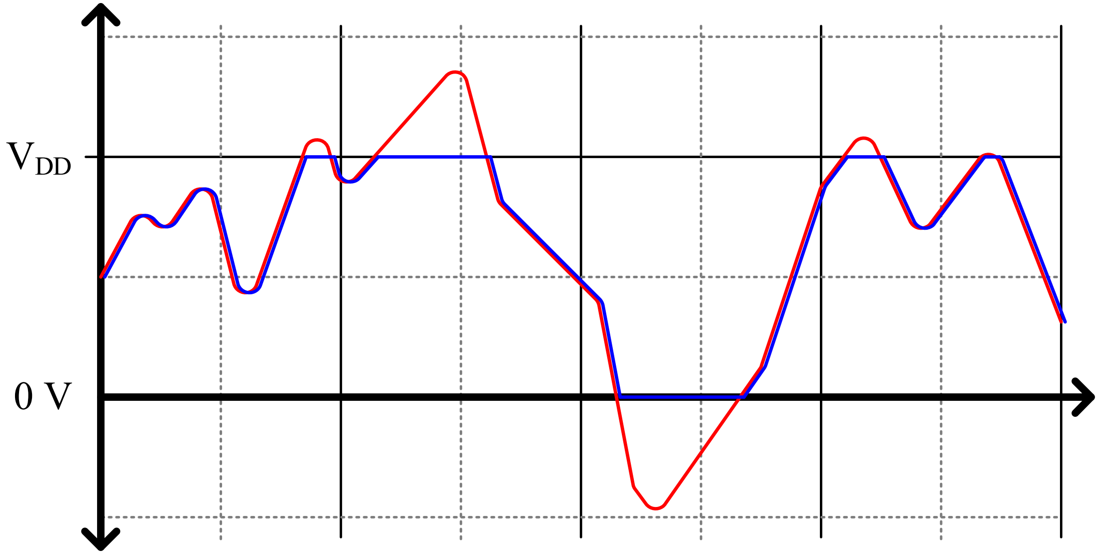
 https://everycircuit.com/circuit/6706811834728448

Signal Clamp (Limiter)

- Signal voltages outside the rails can damage some components
- Diodes can limit such signals
  - Called “clamping” or “protection” diodes
  - Like a pressure release valve



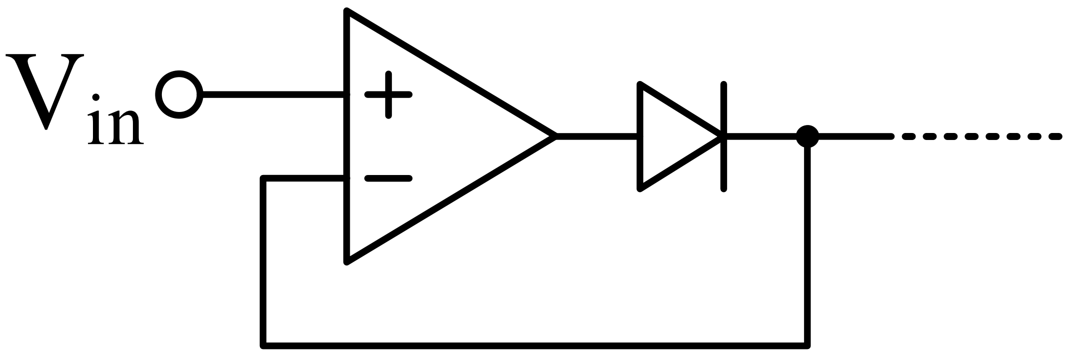




 https://everycircuit.com/circuit/5867101008691200

“Ideal” Diode

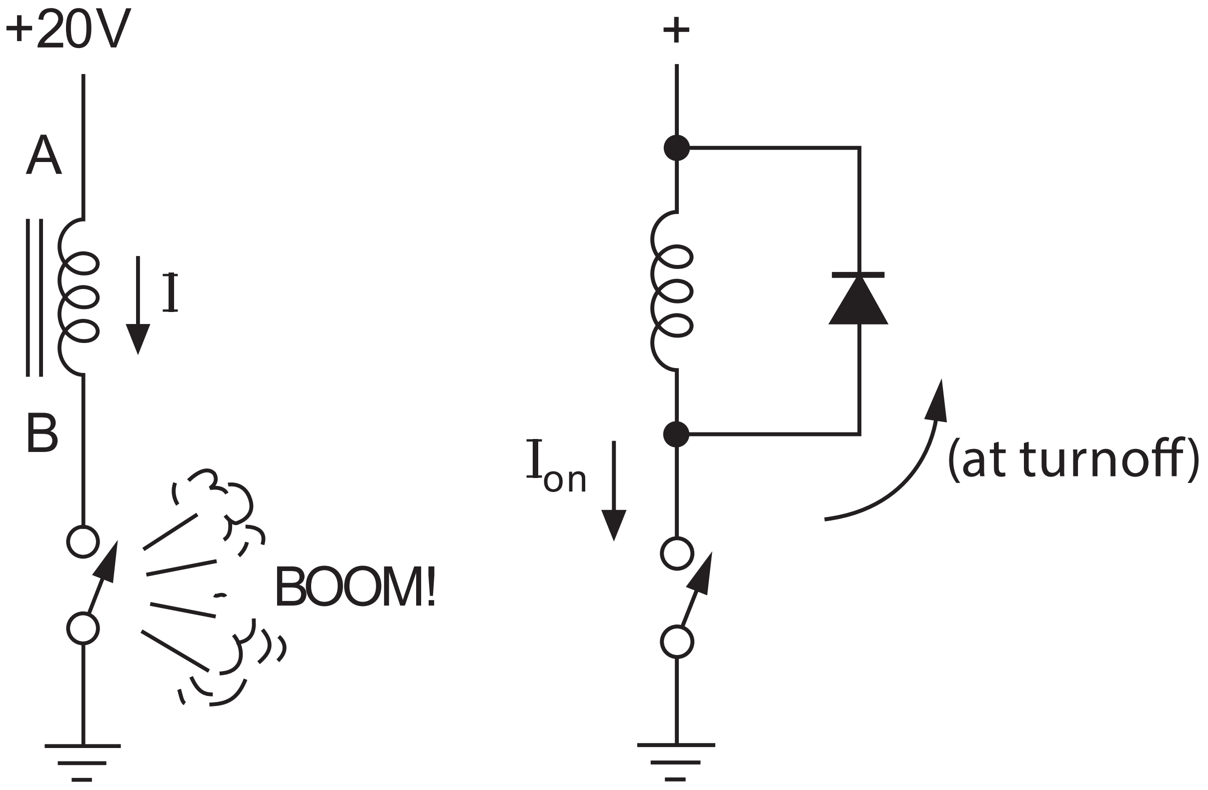
- Feedback “cancels” opamp forward voltage drop



 https://everycircuit.com/circuit/4844104827273216

Protection for Inductive Kickback

- Recall how inductors react to dI/dt
- Diode gives safe pathway for magnetic field to discharge



 https://everycircuit.com/circuit/5692448965394432

LTSpice Modeling Challenge

- Create an opamp circuit that drives an LED current I_D based on an input voltage V_{set} and works regardless of variation in V_F
  - Using your new deeper understanding of opamps in closed-loop negative feedback to control I_D
  - Assume you have a precision resistor of your choosing
  - Choose an LED from Digikey
  - See if you can find an opamp model in LTSpice that is appropriate for your application (can it drive the required I_F?)
  - Can you model V_{set} as a potentiometer wiper voltage (hint: ".param")